

CU

Parameters

- PARTIAL_SIZE.
- DATA_WIDTH.
- DATA_SIZE = PARTIAL_SIZE * DATA_WIDTH.
- NUMBER_OF_PARTIALS_PER_OUTPUT = 2.

Input Signals

Signal	Number of bits	Description
clk	1	The clock
rst_n	1	Asynch active low reset
partial_valid	1	When high, means the partial_data going to the input sampler is valid
segment_step	1	When high, means the partial_data going to the input sampler is valid and we are going to process the next segment
phase_change	1	When high, means the partial_data going to the input sampler is valid and if we are in the INITIAL_PARTIALS state we change the state to ACCUMULATED_PARTIALS state and if already in the second state we change the direction of going through the range
next_range	1	When high, means partial_data going to the input sampler is valid and we go back to INITIAL_PARTIALS state, if not already in it, and the data in this range is ready for the output

operation_done	1	When high, means partial_data goint to the input sampler is valid and we go back to INITIAL_PARTIALS state, if not already in it, and the data in this range is ready for the output
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Output Signals

Signal	Number of bits	Description
sample_data	1	High whenever the data going to the input sampler is valid
alu_read_en	1	High whenever in the ACCUMULATED_PARTIALS, low otherwise
Initial_partial_end	1	High whenever in the INITIAL_PARTIALS and phase_change, range_end or operation_end is high, low otherwise
accumulated_partial_end	1	High whenever in the ACCUMULATED_PARTIALS and range_end or operation_end is high, low otherwise
direction_change	1	High whenever phase_change is high
alu_input_sel	1	High when in ACCUMULATED_PARTIALS, low otherwise

FSM

When reset is asserted, we are in INITIAL_PARTIALS.

When in INITIAL_PARTIALS, the only way to go to ACCUMULATED_PARTIALS is that phase_change is high.

When in ACCUMULATED_PARTIALS, the way to go to INITIAL_PARTIALS is when range_end or operation_end is high.